CareerNovacoke



Present by：疯狂星期四

01

INTRODUCTION

Ethical Principles of Artificial Intelligence



Introduction

SDC HACKATHON 2026

XIAMEN UNIVERSITY MALAYSIA

Team Name 疯狂星期四

Selected Domain Domain 1

Selected SDG Track (1–17) SDG 4, SDG 8, SDG 9, SDG 10

Project / Solution Name CareerNovacoke

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PROBLEM STATEMENT



Problem Statement



Career Uncertainty

- Fresh graduates and career changers may not know which careers suit their skills, interests, and experience.
- Workers may struggle to decide their next career move as AI changes the job market.

Lack of Future Job Market Insights

- AI is changing the skills required for many jobs.
- People may not know which careers are growing, which skills are becoming more important, or how their current skills may fit future opportunities.

Limited Career Guidance

- People may lack clear guidance on the skills, qualifications, and learning resources needed for their target careers.
- Employers and educational institutions may also struggle to identify future skill needs and prepare people for changing job demands.

Problem Statement

Impact of the problem

- Workers may face job insecurity or lower incomes when their skills no longer meet market needs.
- Fresh graduates may struggle to find suitable jobs, while employers may find it difficult to recruit skilled workers.
- Skills gaps may slow progress in important industries such as clean energy, healthcare, and smart agriculture.

Why Is a Solution Needed?

Existing career guidance may not provide a complete picture of personal strengths, changing job market trends, and practical learning steps in one place.

We need a platform that helps people explore suitable careers, understand future skill requirements, and plan their next steps in the AI era.

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TECHNICAL ROADMAP



Technical Approach

Technology Stack

- Frontend: HTML, CSS, JavaScript, and Flask templates
- Backend: Python and Flask
- Career Data: Structured career information stored in Python data structures
- Development & Hosting: GitHub and Replit

System Workflow:

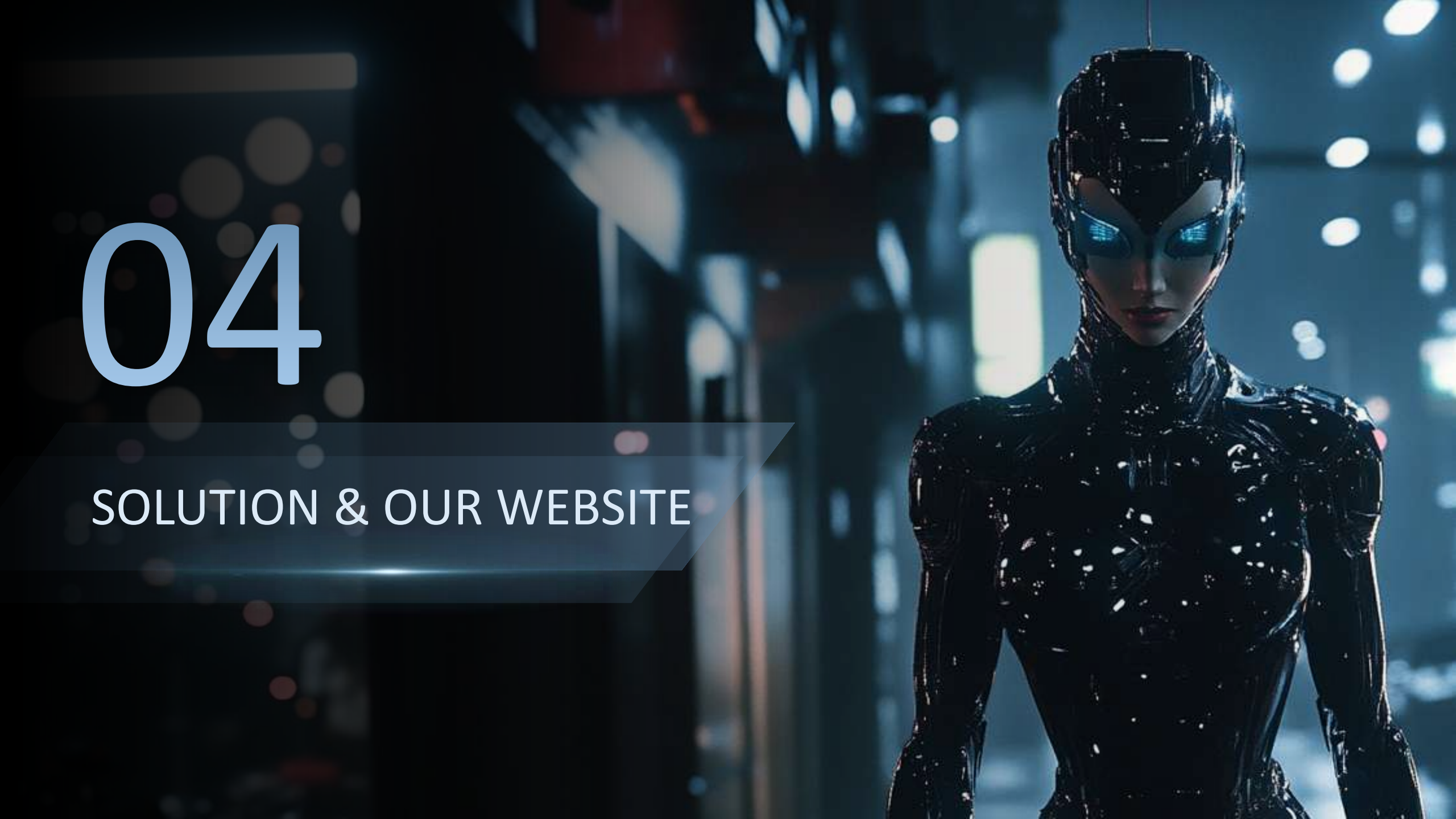
1. Users complete a career questionnaire.
2. The system compares their answers with predefined career requirements.
3. Matching scores are calculated to rank suitable careers.
4. Users receive career details, skill-gap insights, and suggested learning resources.

Matching & Personalisation

- Rule-based scoring evaluates skills, interests, and other career preferences.
- Learning suggestions are selected based on users' existing skills and target careers.
- Predefined career information provides details about salaries, qualifications, countries, and industry outlooks.

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SOLUTION & OUR WEBSITE



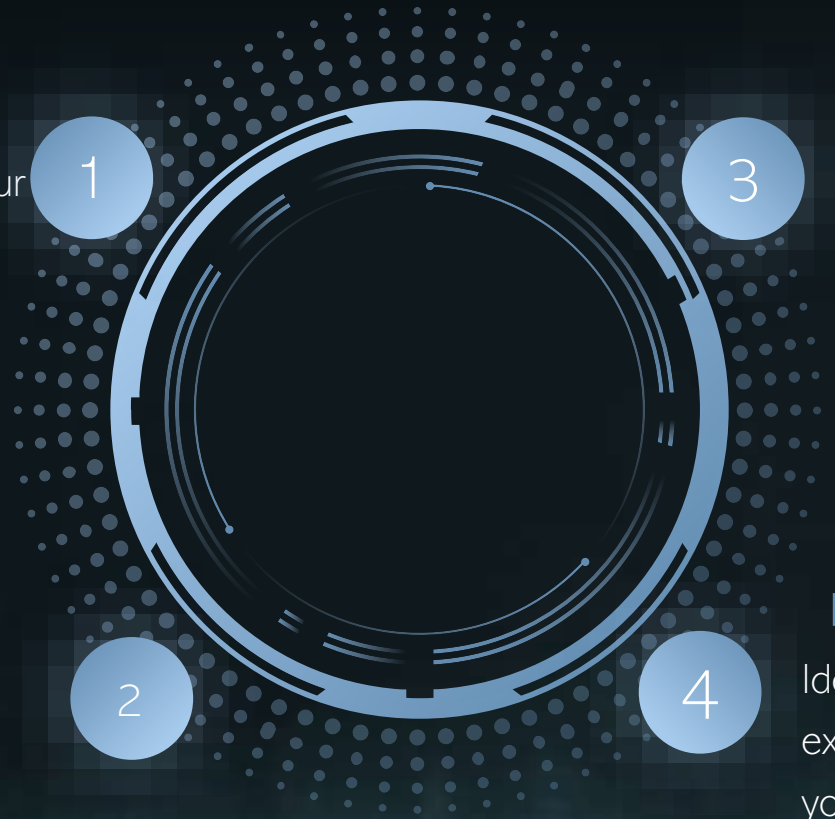
How It Works

1 Discover Yourself

Complete a short questionnaire about your skills, interests, and career preferences.

2 Find Your Career Matches

The system calculates matching scores and ranks careers based on predefined criteria.



3 Explore Career Information

View career descriptions, estimated salary ranges, target countries, certification requirements, and relevant learning resources.

4 Plan Your Next Steps

Identify skills to develop and explore learning suggestions for your chosen career.

How does CareerNovacoke help?

CareerNova is a career guidance platform that helps users discover suitable careers based on their skills, interests, and preferences.

CareerNova brings career matching and career information together in one platform, helping users explore career options and make more informed decisions about their future.



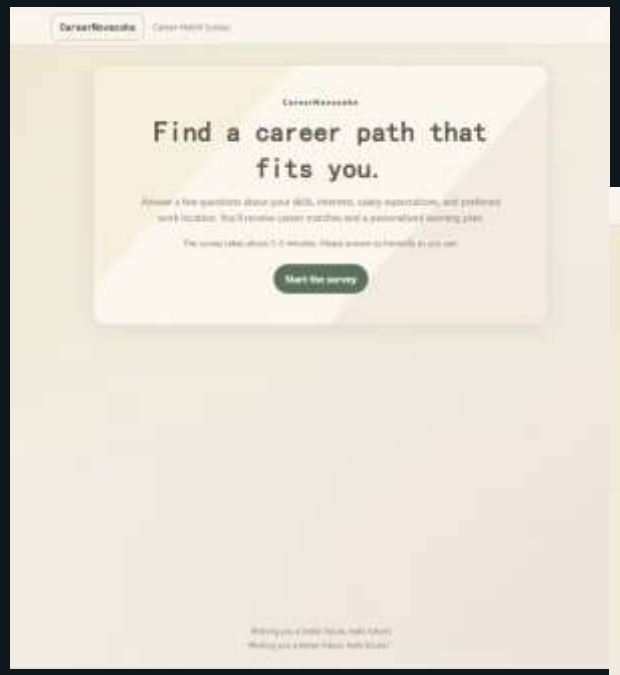
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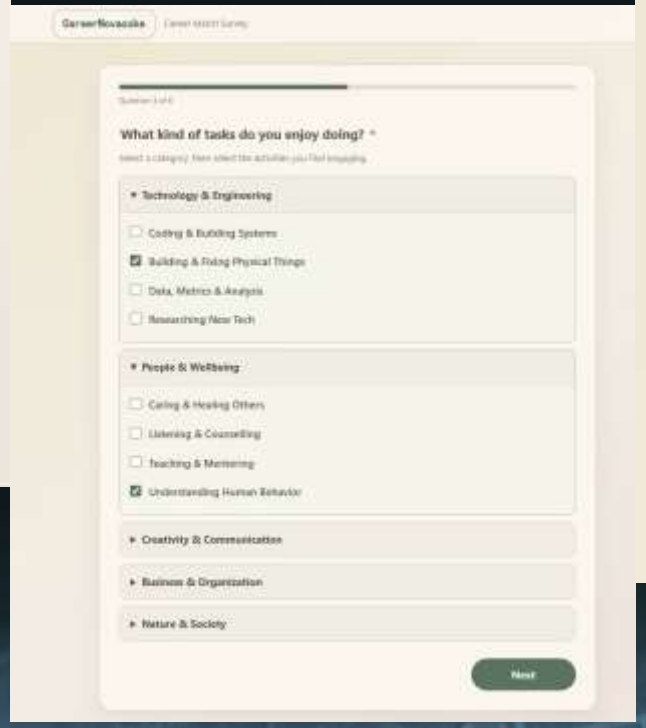
DEMO WEBSITE LINK

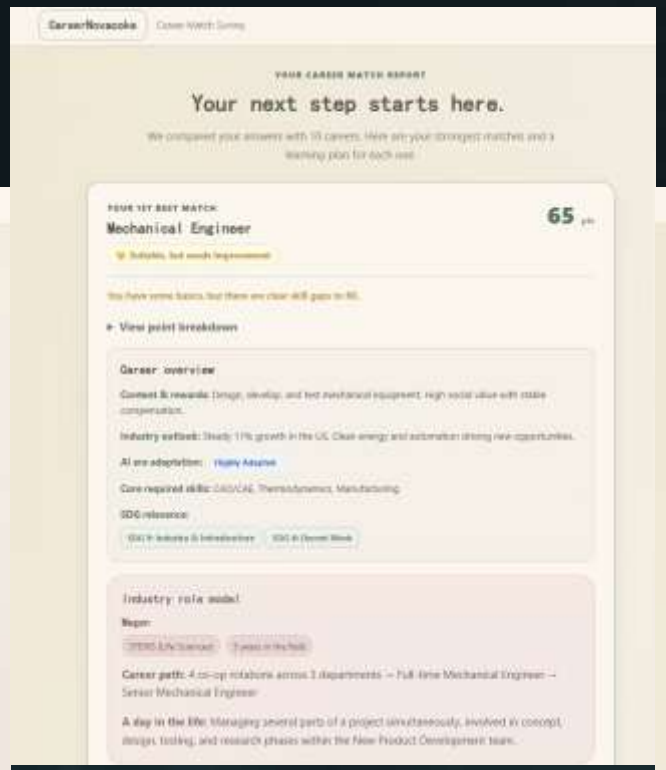
Algorithmic bias and fairness

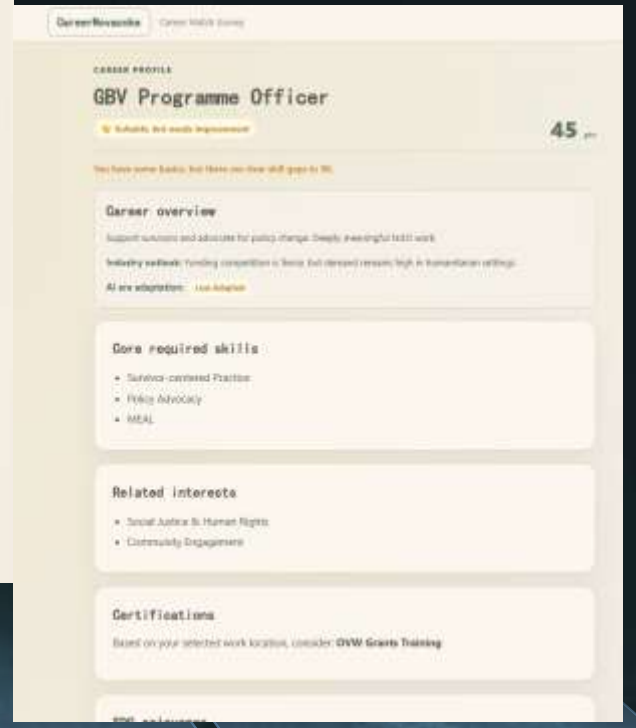
Demo Link

HTTPS://CAREERNOVA-COKE.REPLIT.APP









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TARGET USERS AND USER NEEDS



Target Users

1. College Students & Fresh Graduates：
Especially juniors and seniors facing internships and job hunting, or those wanting to switch majors and explore new career paths.

2. Early-Career Professionals & Career Switchers：
Individuals with 1-5 years of experience who feel the impact of AI, fear being replaced, and urgently need to find new directions.

3. International Students & Returnees:
Cross-border job seekers deciding whether to stay abroad or return home, needing insights into different countries' salary levels, certification requirements, and industry outlooks.

Needs

Eliminate Information Gaps

A "career encyclopedia" offering transparent, real-world information on salaries, skills, certifications, and entry barriers.

Alleviate AI Anxiety

A clear "risk-resilience indicator"—knowing which careers are safe (highly adapted to AI) and which are at risk.

Actionable Guidance

Specific course recommendations, phased learning plans, and visibility into how real people achieved success (Mentor Profiles).

Align with Social Value (SDG)

Understanding a career's positive impact on the world (SDG tags) to find a sense of professional identity.

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EXPECT VALUE AND SOCIAL IMPACT



Expected Value (Value to Individual Users)

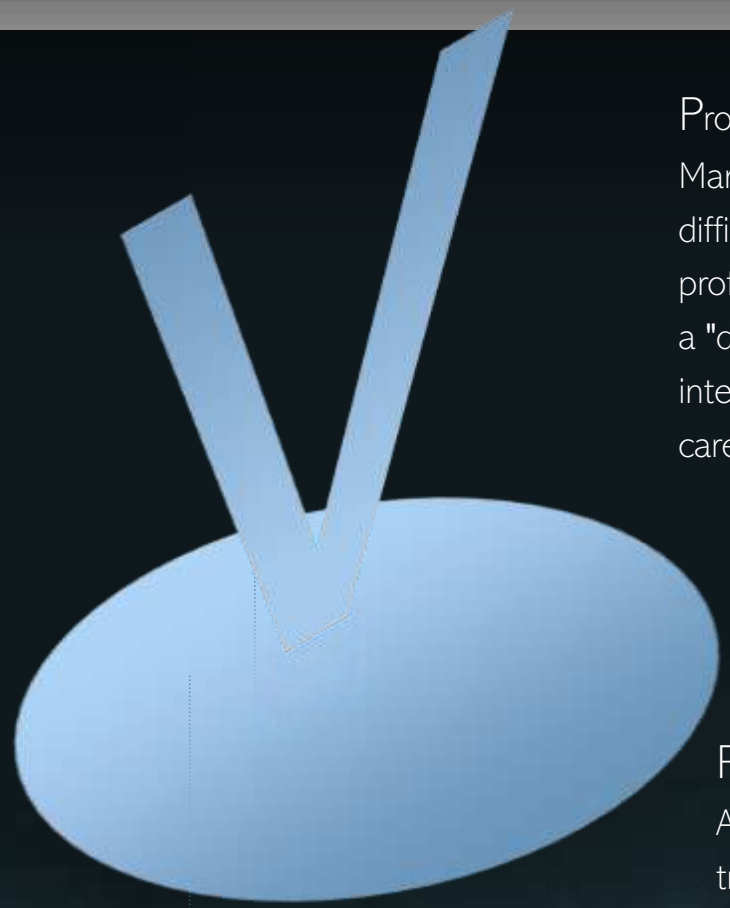
1. Eliminate Information Gaps and Reduce the "Career Trial-and-Error Cost"
Many young people (especially college students or career changers) are completely unaware of the existence of certain careers or the skills required. Through precise questionnaires and detailed profiles (including salary, countries, required skills, and learning resources), our platform transforms the vague goal of "I want to find a good job" into a clear action plan: "I need to learn Python and get a specific certificate." This saves users significant time and money.

2. Alleviate "Career Anxiety in the AI Era"
This is the biggest highlight of our project. Everyone is afraid of being replaced by ChatGPT, but our platform assigns a clear ai_match label (Perfectly Adapted / Highly Adapted / Less Adapted) to each career. This prevents blind panic and shows users which directions are safe to pursue in the AI era (e.g., AI Engineers are "Perfectly Adapted," while Psychologists are "Less Adapted").

3. Upgrade from "What Job Fits Me" to "How Do I Get There"
Ordinary tests only provide results, but our system also offers Mentor Profiles and Phased Learning Plans. This allows users to see how real "industry benchmarks" achieved their success, providing strong action guides and emotional value.

Social Impact (Value to the World and Society)

Deeply Bind Individual Career Choices with SDGs (Sustainable Development Goals)
Traditional career planning often only focuses on "money" and "prospects." Our platform connects careers with SDG tags (such as SDG 4 Quality Education, SDG 9 Industry & Infrastructure, and SDG 10 Reduced Inequalities). This not only guides young people toward more socially meaningful work but also makes them realize that their individual career development can resonate with solving global challenges.



Promote Educational Equity and Resource Equality
Many children in remote areas or from ordinary families have difficulty accessing high-quality industry mentors or professional career planning consultations. our platform acts as a "digital public welfare career planner." Anyone with an internet connection can access high-quality, globally-minded career matching and growth resources.

Facilitate the Transition of the Workforce Structure into the "AI Era"
At the societal level, our platform is effectively facilitating a "smooth transition." It not only tells users which careers are good but also how to leverage or adapt to AI. This helps reduce structural unemployment caused by the technological revolution and helps society cultivate more forward-looking, interdisciplinary talent.

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CURRENT LIMITATIONS,AREAS FOR IMPROVEMENT,AND FUTURE PLAN



Current Limitations, Areas for Improvement, and Future Plans

Data Dimensions & Diversity

Current Limitation: Only 10 careers are included, which cannot cover all industries, emerging fields, or non-traditional career paths (e.g., freelance or gig economy).

Improvement: Expand the career database and add localization.

Future Plan: Collaborate with industry experts and HR professionals to continuously update the database, ensuring salary data and certification requirements are accurate for different countries and regions.

Real AI & Personalization

Current Limitation: The "AI adaptation" tags are currently static, human-assigned labels, and the personalized explanations are template-based rather than generated by a real Large Language Model (LLM).

Improvement: Integrate an LLM API for dynamic content generation.

Future Plan: Connect to OpenAI or Anthropic APIs to generate truly personalized career advice based on users' open-ended responses, and use AI to dynamically summarize real-time industry trends.

User Experience & Interface

Current Limitation: The survey interaction is relatively simple (only single-choice and grouped multi-select), and the results page presents a massive amount of information at once (salary, certifications, mentors, learning plans, etc.), which can easily lead to "cognitive overload" for users.

Improvement: Introduce progressive disclosure and conversational interaction.

Future Plan: Integrate conversational AI to guide users through the test like a real conversation; change the results page to step-by-step unlocking (e.g., first the match score, then the mentor, and finally the action plan), and add a "one-click share" poster generation feature for social media.

Dynamic Content & Maintenance Mechanism

Current Limitation: Currently, career, salary, and certification data are statically hardcoded in the code. This means every industry update requires modifying the code and redeploying, resulting in extremely high maintenance costs.

Improvement: Establish a Backend Content Management System (CMS) and data scraping mechanisms.

Future Plan: Develop a dedicated operational backend so non-technical staff can easily add new careers and update salary ranges at any time; simultaneously, integrate APIs or web scrapers from job platforms to automatically update industry data (such as emerging roles and salary trends).

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MARKET EXPANSION PARTNERMODELS AND BUSINESS MODEL



Market Expansion

The first step

Target "Specific Anxious Groups":

Target Audience: College juniors and seniors facing job hunting, and career starters looking to switch fields but feeling anxious in the AI era ,especially those transitioning from humanities to STEM, or traditional roles to tech.

The second step

Regional Radiation & Localization：

Leverage the "Country Options" feature in our code to run targeted marketing for "International Students/Returnees." This group is often torn between "which country to develop in" and "whether their major is competitive in the local job market."

Launch "Localization" modules. For example, "What certifications do people need to be an AI Engineer in Malaysia?" This directly addresses the pain points of niche audiences.

The third step: Globalization & Full-Industry Coverage :

As our career database expands from 10 to 100 or 1000, we can launch bilingual or even multilingual versions to enter the international market. Because "confused young people" and "AI anxiety" are global phenomena.

Partnership Models

1.Collaborate with Universities/Educational Institutions
Start by partnering with the career centers or psychology departments of a few universities. Offer them free access to the system in exchange for real user feedback.

2.Collaborate with Industry Communities/Content Creators
Our career library contains many mentor profiles. We can find bloggers such as "AI Engineers" and "Overseas Psychologists" on platforms like Xiaohongshu, Bilibili, and LinkedIn. Invite them to join as "Platform-Certified Mentors." They bring their own traffic and attract their followers to take the assessment.

3. Collaborate with NGOs/Non-Profit Organizations to enhance credibility
Reach out to organizations such as UN youth programs or environmental NGOs. Our"SDG + Career" integration is an excellent entry point. Co-host events such as "Green Career Planning Seminars" or "Youth AI Adaptability Bootcamps." This can secure financial support while expanding our social impact.

Business Model

Basic Assessment (Free)

Users can take the survey and view their matching results and AI adaptability for free.
Purpose: Attract a massive user base and build traffic.

Deep Value-Added Services (charged)

Charge individual users for advanced features. For example, access to a more detailed "phased learning plan," downloadable PDF reports, or 1-on-1 consultations with real industry mentors.

Purpose: Generate cash flow through micro-payments or subscriptions.

Institutional Partnerships

Sell the system to university career centers as a student counseling tool, or help enterprises accurately recruit talent.

Purpose: Capture the core profit margin (bulk accounts and service fees).

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CONCLUSION

Vision and value of CareerNova



Vision & Value

Our project--CareerNova is a socially-driven career navigation platform designed for the AI era:

Vision:

Our vision is to empower every career choice to not only be resilient in the AI era but also resonate with the sustainable development of human society.

Core Value:

For Individuals: Eliminate information gaps and AI anxiety. We provide precise matching, forward-looking "AI adaptability" tags, and actionable learning paths with real mentors.
For Society: Promote educational equity and sustainable development. By linking careers with SDGs and digitalizing career counseling, we guide youth toward meaningful roles and break resource barriers.
For the Future: Facilitate a smooth workforce transition into the AI era by cultivating irreplaceable, interdisciplinary talent.

Thank you for listening

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